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Physics  
Assignment - 3  
Chapter 3 : Kinematics

classmate

Date  
Page

| 1. | Sl. No | Differentiating Property | Distance                                                                                                  | Displacement                                                                                    |
|----|--------|--------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|    | 1.     | Definition               | The complete length of the path between any two points is called distance                                 | Displacement is the direct length between any two points                                        |
|    | 2.     | Denotation               | d                                                                                                         | s                                                                                               |
|    | 3.     | Direction consideration  | To calculate distance, the direction is not considered                                                    | To calculate displacement, the direction is taken into consideration                            |
|    | 4.     | Quantity                 | Distance is a scalar quantity as it only depends upon the magnitude and not the direction                 | Displacement is a vector quantity as it depends upon both magnitude and direction               |
|    | 5.     | Route information        | Distance gives the detailed route information that is followed while travelling from one point to another | As displacement refers to the shortest path, it does not give complete information on the route |
|    | 6.     | Formula                  | speed $\times$ time                                                                                       | Velocity $\times$ time                                                                          |

|    |                                  |                                                                            |                                                                                                    |
|----|----------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 7  | Possible values                  | The distance can only have positive values                                 | Displacement can be positive, negative and zero                                                    |
| 8  | Indication                       | Distance is not indicated with an arrow                                    | Displacement is always indicated with an arrow                                                     |
| 9  | Measurement in Non-straight path | The distance can be measured along a non-straight path                     | Displacement can only be measured along a straight path                                            |
| 10 | Path dependence                  | Distance depends upon the path i.e. it changes according to the path taken | Displacement does not depend upon the path and it only the initial and final position of the body. |

2. Yes, these two definitions of 'average speed' are different.
- Average speed is defined as the total distance travelled divided by the total time elapsed. It is a scalar quantity meaning it has magnitude but no direction.

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

The magnitude of average velocity is the absolute value of the average velocity vector. Average velocity is defined as the total displacement divided by the total



time elapsed. It is a vector quantity, meaning it has both magnitude and direction.

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

Distance  $\rightarrow$  +ve, 0      Displacement  $\rightarrow$  +ve, -ve, 0

$$\text{Distance} \geq \text{Displacement}$$

$$\text{Av. speed} \geq \text{Av. velocity}$$

The key distinction lies in the difference b/w distance (scalar) & displacement (vector).

The magnitude of the average velocity and the average speed will only be the same if the total distance traveled is equal to the magnitude of the displacement. This occurs only when an object travels in a single direction along a straight line.

Magnitude of average velocity

The average velocity ( $\vec{v}_{\text{avg}}$ ) is a vector defined as the total displacement ( $\Delta \vec{x}$ ) divided by the elapsed time ( $\Delta t$ )

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t} = \frac{\vec{x}_{\text{final}} - \vec{x}_{\text{initial}}}{\Delta t}$$

The magnitude of the average velocity,  $|\vec{v}_{\text{avg}}|$ , represents the overall rate of change of position in a specific direction.

## Average Speed

The average speed ( $v_{avg}$ ) is a scalar defined as the total distance traveled divided by the elapsed time

$$v_{avg} = \frac{\text{Total Distance}}{\Delta t}$$

Average speed measures how fast an object is moving on average, regardless of its direction

### Example

Consider a car driving from home to the store & back. The car drives 5 km east to a store and returns taking 30 minutes (0.5 hours) for the round trip.

### Average speed calculation

The car's total distance is 5 km (to the store) + 5 km (back home) = 10 km

$$v_{avg} = \frac{\text{Total Distance}}{\text{Elapsed time}} = \frac{10 \text{ km}}{0.5 \text{ h}} = 20 \text{ km/h}$$

### Magnitude of average velocity calculation

The car's total displacement is zero because its final position (home) is the same as its initial position (home)

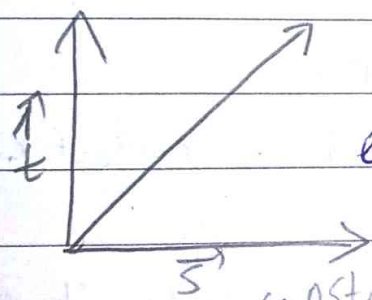
$$|\vec{v}_{avg}| = \left| \frac{\text{Total Displacement}}{\text{Elapsed Time}} \right| = \left| \frac{0 \text{ km}}{0.5 \text{ h}} \right| = 0 \text{ km/h}$$



In this example, the two values are completely different: the average speed is  $20 \text{ km/h}$ , while the magnitude of the average velocity is  $0 \text{ km/h}$ .

3. No, when velocity is constant, the average velocity over any time interval is identical to the instantaneous velocity at any instant. This is because there is no change in speed or direction over time.

$$V_{in} = V_{av} \text{ at constant velocity}$$



In uniform motion object covers equal distances in equal interval of time.

$\therefore$  av. velocity over any interval will be same as constant inst. velocity.

4. Yes, a body can have zero instantaneous velocity but still be accelerating, a ball thrown vertically upward; at its highest point, the ball's velocity momentarily becomes zero, but it continues to accelerate downwards due to gravity.
- $g = 9.8 \text{ m/s}^2$

5. Yes, a body can have a constant speed and still have a varying velocity because speed & velocity are different physical quantities with distinct properties. Speed is a scalar quantity, meaning it is defined only by its magnitude (numerical value). It tells how fast an object moves. Example: a car traveling at 60 km/h.

Velocity is a vector quantity defined by both magnitude & direction. For an object's velocity to be constant, both its speed & its direction of motion must be constant.

Example: Uniform circular motion

Imagine a car driving around a perfect circular track at 30 km/h.

Speed of car is constant at 30 km/h since magnitude of its velocity isn't changing.

However, velocity of car is continuously changing since its direction of motion is constantly changing as it follows the curve of the track.

6. Yes, an object can have an eastward velocity & a westward acceleration simultaneously. This happens when the object is decelerating.





When an object has a velocity in east and an acceleration in opposite direction west it experience a decrease in its velocity. This situation is commonly referred to as deceleration. The object slow down cause the acceleration is acting against its motion.

example: car moving east that applies brakes. Car has initial eastward velocity, but when brakes applied experience westward acceleration (deceleration) as a result car continues to move east for a while but gradually slows down until it stops. So it concludes that it is possible for an object to have eastward velocity while experiencing westward acceleration. The object will decelerate until it either stops or changes direction.

7. Average velocity of the car in x-direction for the first 4 s  $V_{av,x} = 6.25 \text{ m/s}$

$$\begin{aligned} \text{Distance travelled by car in 4s} &= \text{velocity} \times \text{time} \\ &= 6.25 \text{ m/s} \times 4 \text{ s} \\ &= 25 \text{ m} \end{aligned}$$

Car travels 25 meters in 4 seconds.

8. Distance from the light

$$x(t) = bt^2 - ct^3$$

$$b = 2.40 \text{ m/s}^2 \quad c = 0.120 \text{ m/s}^3$$

$$V_{\text{avg}} = \frac{\Delta x}{\Delta t}$$

a) Distance from light  $\Rightarrow x(t) = bt^2 - ct^3$

At  $t = 0$

$$x(0) = b(0)^2 - c(0)^3 = 0 \text{ m}$$

At  $t = 10 \text{ s}$

$$\begin{aligned} x(10) &= 2.4 \times 10^2 - 0.120 \times 10^3 \\ &= 240 - 120 \\ &= 120 \text{ m} \end{aligned}$$

$$\begin{aligned} V_{\text{avg}} &= \frac{dx}{dt} = \frac{120 - 0}{10 - 0} \\ &= \frac{120}{10} \\ &= 12 \text{ m/s} \end{aligned}$$

Average velocity of car for time interval  $t = 0 \text{ s}$  and  $t = 10 \text{ s}$  is  $V_{\text{avg}} = 12 \text{ m/s}$

$$\begin{aligned} \text{b) } V_{\text{inst}} &= \frac{dx}{dt} = \frac{d(bt^2 - ct^3)}{dt} \\ &= 2bt - 3ct^2 \end{aligned}$$



At  $t = 5\text{ s}$

$$V_{\text{inst}} = 2 \times 2.4 \times 5 - 3 \times 0.12 \times 5^2 \\ = 24 - 9 = 15 \text{ m/s}$$

At  $t = 10\text{ s}$

$$V_{\text{inst}} = 2 \times 2.4 \times 10 - 3 \times 0.12 \times 10^2 \\ = 48 - 36 = 12 \text{ m/s}$$

Instantaneous velocity at the time interval  $t = 0$ ,  $t = 5\text{ s}$ ,  $t = 10\text{ s}$  are  $V_{\text{inst}}$  at  $t = 0 \Rightarrow 0 \text{ m/s}$ ,  $V_{\text{inst}}$  at  $t = 5\text{ s} = 15 \text{ m/s}$ ,  $V_{\text{inst}}$  at  $t = 10\text{ s} = 12 \text{ m/s}$ .

c) Given that the car is at rest

so  $V_x = 0 \text{ m/s}$

Since  $V_x = 2bt - 3ct^2$

$$2bt - 3ct^2 = 0$$

$$2bt = 3ct^2$$

$$t = \frac{2b}{3c} = \frac{2 \times 2.4}{3 \times 0.12}$$

$$= \frac{4.8}{0.36}$$

$$= \frac{480}{36}$$

$$= 13.33 \text{ s}$$

The car will be at rest after  $t = 13.33\text{ s}$  again

9. initial speed  $u = 105 \text{ Km/h}$  Given  
 $= 105 \times 10^3 / 3600 \text{ m/s}$   
 $= 29.17 \text{ m/s}$

$$V^2 = u^2 + 2as$$

~~$$0 = (29.17)^2 + 2a$$~~

$V = 0 \text{ m/s}$  as the person is brought to a stop

$$0 = (29.17)^2 + 2(-250) s$$

$$500s = 850.89$$

$$s = \frac{850.89}{500}$$

$$= 1.70 \text{ m}$$

max acceleration

$$a = -250 \text{ m/s}^2$$

-ve because of deceleration

$\therefore$  The air bag must stop the person over a distance of at least 1.70 m for the person to survive the crash.

10. initial speed  $v_i = 2.0 \text{ m/s}$

final speed  $v_f = 1.3 \text{ m/s}$

Distance  $x = 2.0 \text{ cm} = 2 \times 10^{-2} \text{ m}$

Acceleration  $a = ?$

Time  $t = ?$

a)  $v_f^2 = v_i^2 + 2ax$

$$1.3^2 = 2^2 + 2a(0.02)$$

$$\frac{1.69 - 4}{0.04} = a$$

$$a = -2.31 = -57.75 \text{ m/s}^2$$

$$\frac{0.04}{0.04} \approx -58 \text{ m/s}^2$$



$$\begin{aligned}
 \text{b) } v_f &= v_i + at \\
 t &= \frac{v_f - v_i}{a} = \frac{13 - 2.0}{-58} \\
 &= \frac{-0.7}{-58} = 0.012 \text{ s} \\
 &= 12 \text{ ms}
 \end{aligned}$$

$$\begin{aligned}
 \text{11. a) } s &= ut + \frac{1}{2}at^2 \\
 0 &= 3u(10) + \frac{1}{2}(9.8)10^2 \quad \left( \begin{array}{l} (u = 3u) \\ t = 10 \text{ s} \\ a = 9.8 \text{ m/s}^2 \end{array} \right) \\
 u &= \frac{490}{3} = 16.33 \text{ m/s}
 \end{aligned}$$

Time taken by slower stone

$$\Rightarrow s = ut + \frac{1}{2}at^2$$

$$0 = 16.33t + \frac{1}{2}9.8t^2$$

$$4.9t^2 + 16.33t = 0$$

$$4.9t^2 = -16.33t$$

$$t = \frac{+16.33}{4.9} = 3.33 \text{ s}$$

The time of flight for the slower stone is 3.33 s

b)  $2as = v^2 - u^2$       velocity of stone at max height = 0  
 $-2gh = 0 - u^2$        $\therefore v = 0$   
 $h = \frac{u^2}{2g}$

For, slower stone, max height is H & initial velocity is u  $\therefore H = \frac{u^2}{2g}$

$$h = \frac{(3u)^2}{2g}$$

$$= \frac{9u^2}{2g}$$

$$= 9H \quad \left( H = \frac{u^2}{2g} \right)$$

$\therefore$  max. height reached by faster stone is nine times the height of slower stone.



12.

$$r = [4.0 \text{ cm} + (2.5 \text{ cm/s}^2)t^2]\hat{i} + (5.0 \text{ cm/s})t\hat{j}$$

a)

At  $t=0$ 

$$r = (4 \text{ cm} + (2.5 \text{ cm/s}^2)0)\hat{i} + 5.0(0)\hat{j} \\ = 4\hat{i}$$

At  $t=2$ 

$$r = (4 + (2.5)(4))\hat{i} + 5(2)\hat{j} \\ = (4 + 10)\hat{i} + 10\hat{j} \\ = 14\hat{i} + 10\hat{j}$$

$$\text{displacement} \Rightarrow r_2 - r_1 \\ \Rightarrow (14\hat{i} + 10\hat{j}) - (4\hat{i} - 0\hat{j}) \\ \Rightarrow (14 - 4)\hat{i} + (10 - 0)\hat{j} \\ \Rightarrow 10\hat{i} + 10\hat{j}$$

$$\text{Avg velocity} = \frac{dx}{dt} = \frac{10\hat{i} + 10\hat{j}}{2 - 0} = 5\hat{i} + 5\hat{j}$$

$$\text{magnitude of avg. velocity} \Rightarrow \sqrt{5^2 + 5^2} \\ \Rightarrow \sqrt{2(5^2)} \\ \Rightarrow 5\sqrt{2}$$

$$\Rightarrow 7.07 \text{ cm/s} \\ \approx 7.1 \text{ cm/s}$$

direction of dot's avg. velocity  $\Rightarrow$ 

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\tan^{-1}\left(\frac{5}{5}\right) \Rightarrow \tan^{-1}(1) \Rightarrow 45^\circ$$

$$b) \quad r = (4 \text{ cm} + (2.5 \text{ cm/s}^2)t^2) \hat{i} + (5 \text{ cm/s})t \hat{j}$$

$$v = \frac{dr}{dt} = \frac{d((4 + 2.5t^2) \hat{i} + (5t) \hat{j})}{dt}$$

$$= 5t \hat{i} + 5 \hat{j}$$

at  $t = 0$

$$v = 0 \hat{i} + 5 \hat{j}$$

magnitude of instantaneous velocity at  $t=0$

$$\Rightarrow \sqrt{0^2 + 5^2} = \sqrt{25} = 5 \text{ cm/s}$$

direction of instantaneous velocity at  $t=0$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{5}{0}\right) = \tan^{-1}(\infty) \Rightarrow 90^\circ$$

at  $t = 1 \text{ s}$

$$v = 5 \hat{i} + 5 \hat{j}$$

magnitude of instantaneous velocity at  $t=1$

$$\Rightarrow \sqrt{5^2 + 5^2} = 5\sqrt{2} = 7.07 \text{ cm/s} \approx 7.1 \text{ cm/s}$$

direction of instantaneous velocity at  $t=1$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{5}{5}\right) = \tan^{-1}(1) = 45^\circ$$

at  $t = 2 \text{ s}$

$$v = 10 \hat{i} + 5 \hat{j}$$

magnitude of instantaneous velocity at  $t=2$

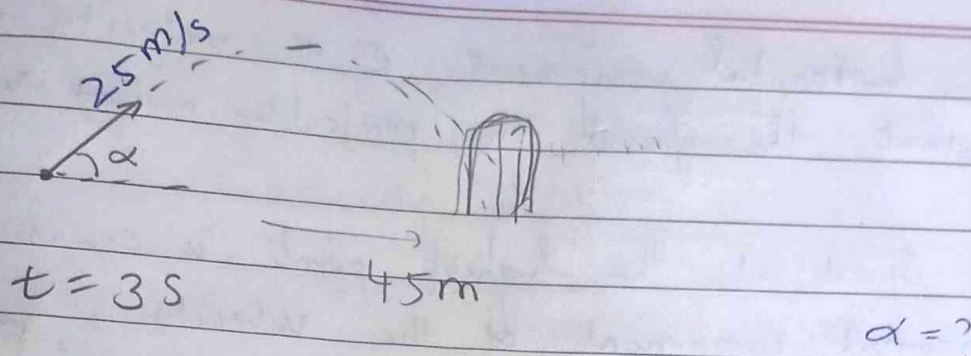
$$\Rightarrow \sqrt{10^2 + 5^2} = \sqrt{125} = 5\sqrt{5} = 11.18 \approx 11.9 \text{ cm/s}$$

direction of instantaneous velocity at  $t=2$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{5}{10}\right) = \tan^{-1}\left(\frac{1}{2}\right) = 26.56^\circ \approx 27^\circ$$



13.



$$\begin{aligned}
 V_{iy} &= 25 \sin \alpha \\
 V_{fy} &= 0 \\
 t &= t_1 + t_2 = 3 \text{ s}
 \end{aligned}$$

- a) The horizontal distance travelled by the water is  $x = (V_0 \cos \alpha) t$  where  $V_0$  is the initial speed,  $\alpha$  is the angle of elevation &  $t$  is the time

Given  $x = 45 \text{ m}$      $V_0 = 25 \text{ m/s}$      $t = 3 \text{ s}$

~~$\cos \alpha =$~~      $x = (V_0 \cos \alpha) t$

$$45 = 25 \cos \alpha \cdot 3$$

$$\cos \alpha = \frac{45}{25 \times 3} = \frac{15}{25} = \frac{3}{5}$$

$$\alpha = \cos^{-1}\left(\frac{3}{5}\right) = 53.13^\circ$$

- b) Speed & acceleration at the highest point  
 At the highest point of its trajectory, the vertical component of the water's velocity is zero  
 $V_y = 0$

The horizontal component of the velocity remains constant throughout the projectile motion  $V_x = V_0 \cos \alpha$

The speed at the highest point is equal to the horizontal component of the velocity;

$$v = V_x = \cancel{25} V_0 \cos \alpha = 25 \cos (53.13^\circ) \\ = 25 \times 0.6 = 15 \text{ m/s}$$

The acceleration of water at the highest point is solely due to gravity, acting downwards so  $a = g = 9.8 \text{ m/s}^2$

c) The height above the ground where the water strikes the building is determined by the vertical displacement  $y = (V_0 \sin \alpha) t - \frac{1}{2} g t^2$

$$y = (25 \sin (53.13^\circ)) 3 - \frac{1}{2} 9.8 (3)^2 \\ = 59.9 - 44.1 \\ = 15.8 \text{ m}$$

The vertical component of velocity just before impact

$$\Rightarrow V_y = V_0 \sin \alpha - g t \\ = 25 \sin (53.13^\circ) - 9.8 \times 3 \\ = 19.9 - 29.4 \approx 20 - 29.4 \\ = -9.4 \text{ m/s}$$

The speed just before impact is calculated using pythagorean theorem,

$$V = \sqrt{V_x^2 + V_y^2} = \sqrt{15^2 + (9.4)^2} = \sqrt{225 + 88.35} = \sqrt{313.35} \\ = 17.7 \text{ m/s}$$



- 14: a) The initial vertical velocity of rock relative to the ground is determined by the balloon's downward velocity which is 20 m/s  
The time of flight of rock = 6 s

$$\begin{aligned} y &= v_{0y}t + \frac{1}{2}at^2 \\ &= 20(6) + \frac{1}{2}(9.8)6^2 \\ &= 120 + 176.4 \\ &= 296.4 \text{ m} = 296 \text{ m} \end{aligned}$$

$$\begin{aligned} v_{0y} &= 20 \text{ m/s} \\ t &= 6 \text{ s} \\ a &= 9.8 \text{ m/s}^2 \end{aligned}$$

height of balloon when the rock was thrown out  $\Rightarrow 296.4 \text{ m}$

b) change in height of balloon  
 $\Rightarrow \Delta y = v_{\text{balloon}} t = -20 \times 6 = -120 \text{ m}$

The height of balloon when the rock hits the ground is the initial height minus the distance the balloon traveled during this time

Height of balloon when rock hits ground

$$\Rightarrow \text{initial height} - \Delta y$$

$$\Rightarrow 296.4 - 120 = 176.4 \text{ m} = 176 \text{ m}$$

c) Rock's horizontal displacement

$$x_{\text{rock}} = v_x t = 15 \times 6 = 90 \text{ m}$$

Rock's vertical displacement

$$\Delta y_{\text{rel}} = \frac{1}{2}gt^2 = \frac{1}{2}9.8 \cdot 6^2 = 176$$

Distance between rock & basket

$$d = \sqrt{x^2 + y^2}$$

$$= \sqrt{90^2 + (176.4)^2}$$

$$= \sqrt{8100 + (176.4)^2} = \sqrt{39216.98} = 198 \text{ m}$$

The rock is 198 m from the basket

d) i) Observer at rest in the basket

Horizontal velocity = 15 m/s since there is no horizontal acceleration, this velocity remain constant

Vertical velocity

$$V_y = V_0 + at = 0 + (-9.8 \text{ m/s}^2) 6 \text{ s} \\ = -58.8 \text{ m/s}$$

$$\boxed{V_x = 15 \text{ m/s} \quad V_y = -58.8 \text{ m/s}}$$

ii) Observer at rest on ground

$V_x = 15 \text{ m/s}$  since basket is not moving

$$V_y = V_0 + at = -20 - 58.8 = -78.8 \text{ m/s}$$

$$\boxed{V_x = 15 \text{ m/s} \quad V_y = -78.8 \text{ m/s}}$$



15. The relative velocity of canoe w.r.t the earth moving south east  $= 0.40 \text{ m/s}$   
The relative velocity of the river  $= 0.5 \text{ m/s}$

x-component of canoe velocity relative to earth

$$\begin{aligned}V_{ex} &= V_e \cos 45^\circ \\&= 0.40 \cos 45^\circ \\&= 0.2828 = 0.283 \text{ m/s}\end{aligned}$$

y-component of canoe velocity relative to earth

$$V_{ey} = V_e \cos 45^\circ = 0.283 \text{ m/s}$$

Canoe velocity relative to river

$$V_{crx} = 0.283 \text{ m/s} - 0.5 \text{ m/s} = 0.217 \text{ m/s}$$

$$V_{cry} = 0.283 \text{ m/s} - 0 \text{ m/s} = 0.283 \text{ m/s}$$

The net velocity of canoe relative to the river

$$\begin{aligned}V &= \sqrt{(V_{crx})^2 + (V_{cry})^2} = \sqrt{(0.217)^2 + (0.283)^2} \\&= \sqrt{0.05 + 0.08} \\&= \sqrt{0.13} \\&= 0.36 \text{ m/s}\end{aligned}$$

Direction of velocity  $\Rightarrow \theta = \tan^{-1} \left( \frac{V_{cry}}{V_{crx}} \right)$

$$\Rightarrow \tan^{-1} \left( \frac{0.283}{0.217} \right)$$

$$= 52.52^\circ \text{ in South west}$$